A Phishing Detection System based on Machine Learning

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***Abstract*— As the Internet has become an essential part of human beings' lives, a growing number of people are enjoying the convenience brought by the Internet, while more are attacks coming from on the dark side of the Internet. Based on some weaknesses of human nature, hackers have designed confusing phishing pages to entice web viewers to proactively expose their privacy, sensitive information.**

In this article, we propose a URL-based detection system - combining the URL of the web page URL and the URL of the web page source code as features, import Levenshtein Distance as the algorithm for calculating the similarity of strings, and supplemented by the machine learning architecture. Due to the great accuracy in small sample numbers and binary classification, we implement Support-vector machine to be the machine learning algorithm model in our system. The system is designed to provide high accuracy and low false positive rate detection results for unknown phishing pages.

I. Introduction

With the develop of network technology, in addition to allowing network users to have a better and more comfortable experience, the changing nature of the network platform also makes cybercriminals have a variety of attacks. Among them, there is a long-standing and well-known phishing attack in the history of the Internet, that is, phishing attack. Phishing is a criminal cyberattack that steal victim's privacy information via deceptive e-mails or counterfeit webpages [1]. The word phishing is a neologism created by combining phone and fishing [2]. As the name suggests, phishing attack's target are people's desires. They may be curious to be satisfied or may be opportunistic greed.

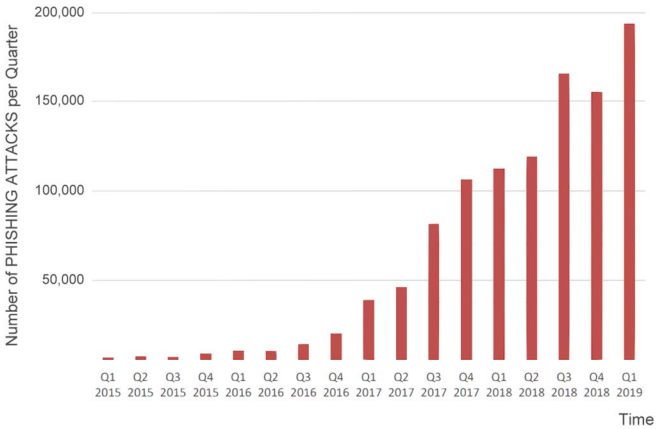

Figure 1. Phishing Activity Trends Report 1st Quarter 2019

According to the report from the Anti-Phishing Working Group [3], the number of phishing attacks increase rapidly, which not only shows the damage caused by phishing attacks is getting more and more serious, but also demonstrate the urgency and importance of phishing detection nowadays. Fig.1 shows how the phishing attack increased and the increasing trend recent years.

Based on the fact that many phishing pages have its URL similar to those legal and popular websites, this study proposes a URL-based detection system that combines the URL of the web page and the URLs contained in the web page source code to define plurality of sets of features which are provided for machine learning to train the detection model for automatically detection whether the webpage is a phishing webpage or not when facing an unknown suspicious webpage.

In this study, we propose an approach based on machine learning Support Vector Machine to help users to classify an unknown webpage. Most of Tradition way use blacklist-based matching to avoid user access the malicious website, however, it not only takes the high cost of labor for human verification but also cannot detect zero-day phishing attack due to the delay before these malicious phishing webpage being added to the blacklist. Our study is able to reduce the labor cost, and detect whether the webpage accessing is a phishing webpage or not with the help of machine learning .In addition, to improve the accuracy, we merge Fuzzy-logic into machine learning architecture to replace Boolean algorithm. Combining the machine learning system with the fuzzy logic, we obtain a good result in terms of identification accuracy.

Due to the characteristic we mention above, the feature we design to input the machine learning contain the similarity parameters, including the difference between the webpage's URL's domain name and those URLs' domains which are contain in the webpage, and also subdomain and path. Besides, since the phishing pages' life time are generally short, we add the webpage's Time To Live as an feature with reference value.

For the sake of alerting users when access an phishing webpage, we build part of our system in browser extension, including web content extraction and alerting partition. Therefore, whenever users access an webpage, our system is available to remind them promptly.

This paper discusses the current mainstream phishing detection related techniques in the session II. The system architecture will be introduced in session III, and the detailed experimental process is explained in session IV. Finally, the

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conclusions and future research directions are presented in session V.

II. Related Works

As a long-established and low-tech attack method, why are phishing attacks still highly effective? Google and the UC San Diego study report [4] pointed out that the proportion of people caught in various phishing web traps is as high as 45%, which is much higher than expected. Hackers attempt to defraud users of their personal data by fake websites which copy those legitimate web pages (such as Google's website). 14% of them will automatically provide information to hackers. Even the worst phishing websites, 3% will still be hacked. As Jamz Yaneza, TrendMicro's threat research project manager, said, "Many of your connections may be very secure, and the only connection may cause problems is which between the chair and the keyboard."

Like the magic of manipulating human psychology, social engineering usually uses the public's neglected tricks to let victims fall into traps, allowing them to take specific actions or reveal confidential information. The social engineering trap is most commonly used in Web security threats, using human innate emotions such as empathy, compassion, curiosity and fear to entice people to take action that cybercriminals expect. There are many software and hardware devices that can effectively prevent a variety of network threats. However, trusting the nature of others is one of the most vulnerable aspects of the entire protection system.

A. Content based

Content filtering is a fairly common method of phishing detection. The existing papers [5, 6, 7, 8] have proposed a large number of phishing detection systems based on text analysis, which is commonly used in phishing pages, phishing emails, and more commonly The context of the connection, as a basis for detection, is a phishing detection system with considerable detection performance and general use.

CANTINA [6] was developed by Carnegie Mellon University's Usable Privacy and Security Lab, and its full name is Carnegie Mellon Anti-Phishing and Network Analysis Tool, using TF-IDF (Term Frequency-Inverse Document Frequency) technology, one for Weighting techniques commonly used in information retrieval and word exploration. The method is to first take the top five nouns of the TF-IDF of the browsed webpage, and then use Google to search for the five words to compare whether the current Domain appears in the top 30 domains of the search results. For example, suppose you have an eBay website that doesn't know whether it is true or not. Use TF-IDF technology to find the five most frequent words eBay, PayPal, sign, forgot, and help, and then use these five words. The keyword is searched by Google, and then compares whether the Domain of this unknown website appears in the top 30 domains of the search results, some are legitimate websites, and if not, it is a phishing website.

GoldPhish [7] captures images of the page, employs optical character recognition (OCR) technology to convert the image into text , leverages the Google PageRank algorithm to retrieve the top ranked domains from a search engine and compares them with the current page.

However, such detection systems require a large data set as a basis, and also cause language singularity problems due to language limitations. In addition, the recent new phishing webpages also have the tendency to replace text with images to circumvent such phishing detection systems. At this point, it is necessary to rely on image detection [13] and URL Based two other systems of phishing detection systems.

B. URL based

URL Based is another phishing detection system other than text analysis. Because a lot of phishing will use similar URLs for scams, such as "google.com" and "Google.com", PayPal to Paypa1 and other similar concepts of phishing pages. The existing papers [9, 10, 16] mostly use the cut URL as a plurality of segments for string comparison as the detection basis and are supplemented by black and white lists. However, the URLs contained in most web pages, the cut-out fragments are also different from the URLs in this URL, which is easy to cause errors in judgment.

Phishing Webpage Detection Using Weighted URL Tokens for Identity Keywords Retrieval [9] After cutting the URL, the word weight is given as a rating score by the formula. However, if the phishers successfully register a non-existing ccTLD-based domain name using the same second-level domain as the legitimate domain name, the system will produce a false negative result.

Off-the-Hook: An Efficient and Usable Client-Side Phishing Prevention Application [10] uses both URL based and text analysis as a two-valid phishing detection system, but when phishing a web page uses images to replace the text content of the entire web page at the time, the proposed method may not find enough identity keywords.

Phishing-Aware [11] is a phishing detection which apply on fog network with promising technologies of virtualization, in order to reduce the utilization of the end-point. The concept of its feature extraction consider fuzzy-logic to improve its accuracy.

C. Machine learning

In case of the development and popular of Artificial Intelligence, there is an increasing number of phishing detection approach machine learning method to improve the classification and downsize the labor cost.

Off-the-Hook [10] use Gradient Boosting to build the classification model due to its strong ability of selecting the most import features and weight its priority, also its boosting algorithms are known to solve the problem of overfitting, therefore it enable the resulting model having a good generalization ability.

Based on the imbalance number between phishing and legitimate class, “Detecting New Forms of Phishing Attacks” [12] use Random Under-Sampling Boost algorithm which handles the class imbalance problem via randomly undersampling.

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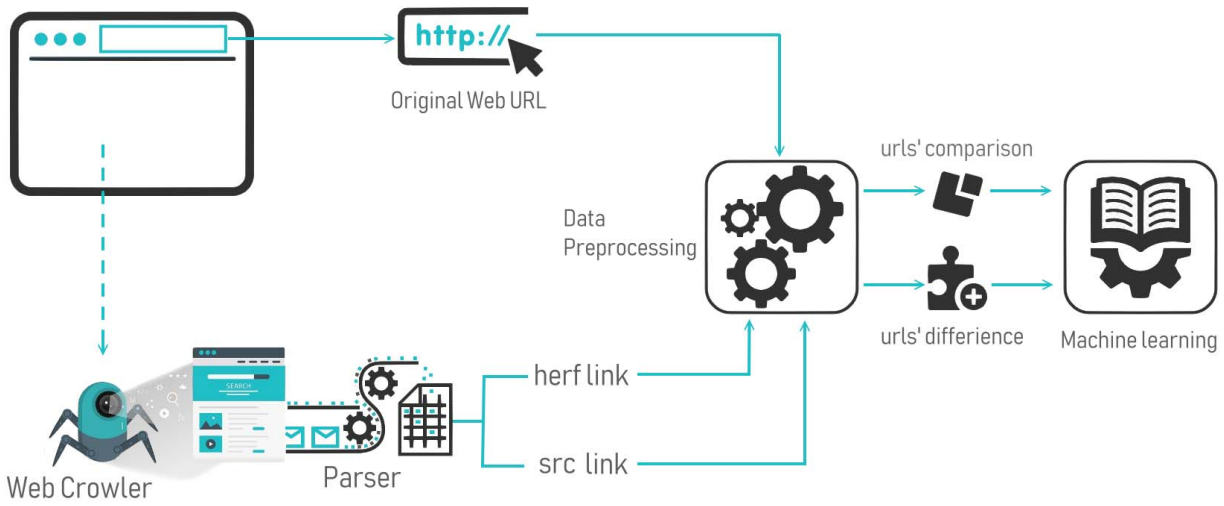

Figure 2. System Architecture

III. Approach

According to observation, a large number of phishing web pages are imitating common websites to deceive web users, entice them enter their own sensitive information on the page in order to steal them. For the sake of preventing the victim from being aware of being deceived, most phishing pages will still transmit the user's information to the imitated webpage, so that the user can browse or access it normally, while the strange process is not detected.

In these situations, phishing web pages must use the API or link to mimics the web page, so the web page source code will contain the URL of the mimicked web page.

Our approach is designed according to the above characteristics. As shown in the figure, the system can be mainly divided into several components, which contains crawler, string processer, and machine learning. The following sections will describe each part in detail.

A. Crawler

First, we will track the main URL of the webpage whether it will be redirected to another destination URL, and also pick the destination URL as the first data.

The second step is scraping the webpage source code with web crawler. Here we use a web testing tool: Selenium [17]: Selenium is a collection of tools designed for the requirement of browser automation, allowing programs directly drive browsers for various website operations, which means Selenium performs a "real browser" to automate the operation on the website. It can directly access the real-time content, including the DOM content modified by JavaScript. Therefore, it is also suitable for A front-end AJAX-enabled website. Here we use Selenium to get real-time content on the web through Selenium to find the URL hidden by the web designer in JavaScript rendering.

URLs embedded in web pages generally use href and src tags, which are different for its usage, so they are separated as different features. Fig.3 shows what characteristic the href tag and src tag act in the webpage's source code.

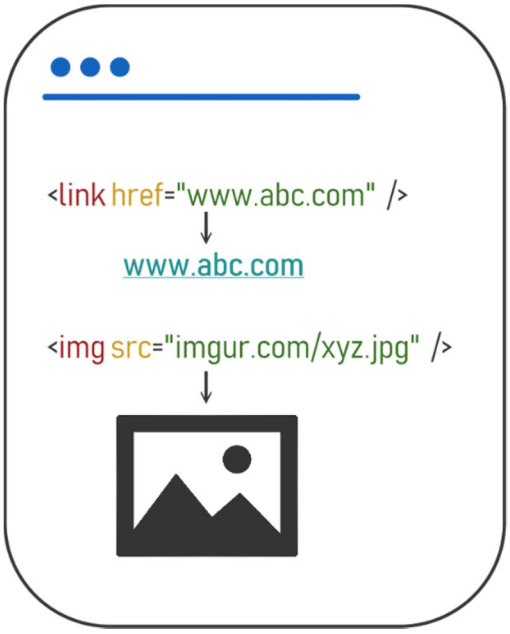

Figure 3. Difference of href link and src link

The Href (Hypertext Reference) is an attribute of the anchor tag, which is also used to identify sections within a document, which specifies the link's destination. The HREF contains two components: the URL, which is the actual link, and the clickable text that appears on the page, called the "anchor text." When reading the href tag in HTML, the browser's parsing of the page will not be suspended, but only as a link.

The src (Source) attribute simply embeds the current resource to the location defined by the current document element. When the browser parses into the src tag, the loading and processing of the page before the compilation and execution of the link file is suspended for presentation to the web page viewer.

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B. *String processor*

In the second division, we will first parse The original web URL into several parts, containing its domain name, subdomain, and paths. Also, the herf links and the src links' will retain its domain name for the following processing.

Next we compare the difference between main url's domain and each contained urls' domain.In this study, in order to define the difference between two strings as features, we use string similarity comparison which its similarity parameter based on edit distance: a way of quantifying how dissimilar two strings are to one another by counting the minimum number of operations required to transform one string into the other. Editing distance can be used in natural language processing. For example, spell checking can determine which are more likely words based on the edit distance of a misspelled word and other correct words.

Different definitions of an edit distance use different sets of string operations. In our study, the edit distance we use is the most common metric: Levenshtein distance, which represent minimum number of single-character edits (insertions, deletions or substitutions) required to change one word into the other. The following process is an example how Levenshtein distance calculate the edit distance.

"kitten" → "sitting":

kitten → **s**itten (substitution of "s" for "k")

sitt**e**n → sitt**i**n (substitution of "i" for "e")

sittin → sitting (insertion of "g" at the end)

Figure 4. Demonstration of Levenshtein distance

We then calculate the similarity between the web page's domain name and those links' domains which are contain in the webpage.According to the above Levenshtein distance, we compare the similarity between the Domain name of the URL of the detected webpage and the Domain name of total URLs contained in the webpage, and distinguish it into three categories: complete consistency, high similarity, and low similarity, then calculate the proportion of each category to all URLs; Subdomain do the same processing: Compare the similarity to the Domain name of all URLs.For the path of the webpage, since it might combine with lots of files' name, we calculate the ratio of high similarity file to domains.

In addition, based on the fact that phishing pages are deleted due to possible reports, most of the survival time is quite short, so we added the time to live of the webpage as another reference.The following table is features we extracts for the machine learning input.

a. Features for Machine learning

	Name	Description	Range
href / src	href_domain_cc	ratio of domain are the same with main url's domain	0~1
	href_domain_high	ratio of domain have high similarity with main url's domain	
	href_domain_low	ratio of domain have low similarity with main url's domain	
	href_subdomain_cc	ratio of domain are the same with main url's subdomain	
	href_subdomain_high	ratio of domain have high similarity with main url's subdomain	
	href_subdomain_low	ratio of domain have low similarity with main url's subdomain	
	href_path_high	ratio of domain have high similarity with main url's path	
	src_domain_cc	ratio of domain are the same with main url's domain	0~1
	src_domain_high	ratio of domain have high similarity with main url's domain	
	src_domain_low	ratio of domain have low similarity with main url's domain	
	src_subdomain_cc	ratio of domain are the same with main url's subdomain	
	src_subdomain_high	ratio of domain have high similarity with main url's subdomain	
	src_subdomain_low	ratio of domain have low similarity with main url's subdomain	
	src_path_high	ratio of domain have high similarity with main url's path	
Time to live		Page exist time	number

b. Comparison of different Machine learning algorithm

Machine learning algorithm	TP rate(%)	FP rate(%)	TN rate(%)	FN rate(%)
SVM	92.6	7.4	93.8	6.2
Decision Tree	82.3	17.7	77.6	22.4
Logistic Regression	58.6	41.4	61.3	38.7

C. *Machine learning*

The machine learning algorithm used in our study is Support Vector Machine. After comparing Decision Tree and Logistic Regression, it is found that SVM can get much better results than other algorithms on small sample training sets.

Since we design our system in python, the machine learning kit we use is scikit-learn[20], a common used machine learning library in python. In our experiment, we try three algorithms with same testing dataset to comparison the performance and detection accuracy. The table below is the result, and the follow equations show how we calculate the rates.

$$\text{TP rate} = \frac{num.of\ phishing\ pages\ detected}{num.of\ all\ phishing\ pages}$$

$$\text{FP rate} = \frac{num.of\ phishing\ pages\ not\ be\ detected}{num.of\ all\ phishing\ pages}$$

$$\text{TN rate} = \frac{num.of\ legal\ pages\ system\ recognized}{num.of\ all\ legal\ pages}$$

$$\text{FN rate} = \frac{num.of\ legal\ pages\ detected\ as\ phishing}{num.of\ all\ legal\ pages}$$

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IV. EXPERIMENT

The experimental set used in this experiment was 5000 phishing pages taken by Phishtank[18], with 10000 normal web pages collected by DMOZ[19]: a multilingual open-content directory of World Wide Web links. PhishTank is a community website for anti-phishing attacks, users can report phishing websites from all over the world and the webpages reported will be trailed by other users via voting. We then label each webpages with 0 for legal webpage and 1 for phishing page. We sliced the experimental set into training data which contain 3000 phishing pages and 5000 legal pages, and testing data contain the remaining.

After complete the model's training, we input the testing data, the accuracy of detecting phishing webpages is 89.3%, while the false positive rate is 6.2%, which means there are 6.2% of legal webpages are judged as phishing webpage.

After observing this phenomenon, we found that some types of phishing pages were obviously not detected by our model. After taking out the samples with the predicted misalignment, we found that there are quite a few phishing websites arranged by using a screenshot of the imitated website, placing the input box at a fixed position to minimize the URL and the text, in order to avoiding URL based or text analysis based phishing detection. In lots of related words, it is also common to discuss the system's circumvention of this type of phishing pages. Most of the literature also mentions that such phishing pages need to be detected by image analysis.

After removing the sample of these specific phishing page and testing with the remaining experimental samples, the accuracy of the judgment returned to 92.6%. The following table is the comparison with another phishing detection mentioned in related works with the same experimental set.

Anti-phishing method	TP rate(%)	FP rate(%)	TN rate(%)	FN rate(%)
Proposed method	92.6	7.4	93.8	6.2
CANTINA[6]	91.78	8.2	80.81	19.19
Weighted URL Tokens[9]	88.43	11.57	87.8	12.2

a. Comparison of different Machine learning algorithm

V. CONCLUSION

In our study, we design an architecture to detect phishing webpage. Due to the fuzzy logic, accuracy is enhanced in our system, which is less mentioned in others' study.

Analysis and research on such phishing pages we mentioned in experiment session will be conducted in the future, hope to avoid more type of phishing webpage. In addition, the front-end part of our system will be design in the future, we decide to alert users when they access an phishing webpage via browser's extension. In the hope that the accuracy of the phishing detection system will be improved, we hope this study can consider for users, in order to against malicious from the dark side of the network.

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